

Energy for escape, field for orbit

Paper companion to the interactive orbits page

Name: _____ Class: _____ Date: _____

Questions about rockets and satellites are solved by choosing between two tools. The potential $V = -GM/r$ has the integral of the changing force already built in, so any question that compares two positions (a maximum height, escape, a change of orbit) is fastest by energy, where the work between two points is $m\Delta V$. A circular orbit asks for a speed at one radius, and that is fixed by the force balance $mv^2/r = GMm/r^2$, so the field strength $g = GM/r^2$ is the tool there. The two meet in the orbital energy $E = -GMm/2r$. Use $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$, $M = 5.97 \times 10^{24} \text{ kg}$ and $R = 6.37 \times 10^6 \text{ m}$ for the Earth, so that $GM = 3.98 \times 10^{14} \text{ m}^3 \text{ s}^{-2}$. Ignore air resistance and the rotation of the Earth unless told otherwise.

1. Choosing the tool

1.1. For each situation below, write “potential or energy” or “field strength and force” in the middle column, and give a reason in the right column. [5]

Situation	Tool	Reason
Speed of a satellite in a circular orbit at 600 km		
Greatest height reached by a rocket fired straight up		
Escape speed from the surface		
Period of a planet's orbit		
Work to move a satellite to a higher orbit		

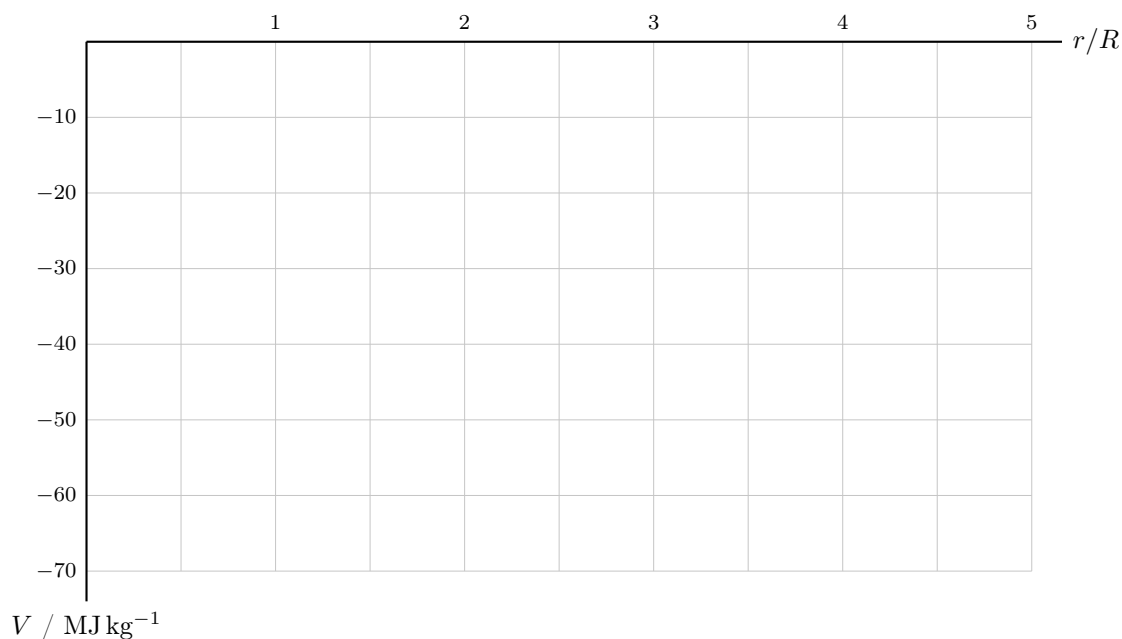
2. Moving between equipotentials

The work done by the engines in moving a mass slowly between two points is $m\Delta V$, the vertical gap between the two points on the graph of V against r .

2.1. Complete the table of gravitational potential at multiples of the radius of the Earth. [2]

r	R	$2R$	$3R$	$4R$
$V / \text{MJ kg}^{-1}$				

2.2. Plot V against r/R on the grid and join the points with a smooth curve. Then draw the tangent to the curve at $r = R$, using the fact that the slope there equals the surface field strength (9.81 N kg^{-1} , which is 62.5 MJ kg^{-1} per Earth radius). State the value of V at which your tangent crosses the axis $V = 0$, and the value of r/R at that point. [5]



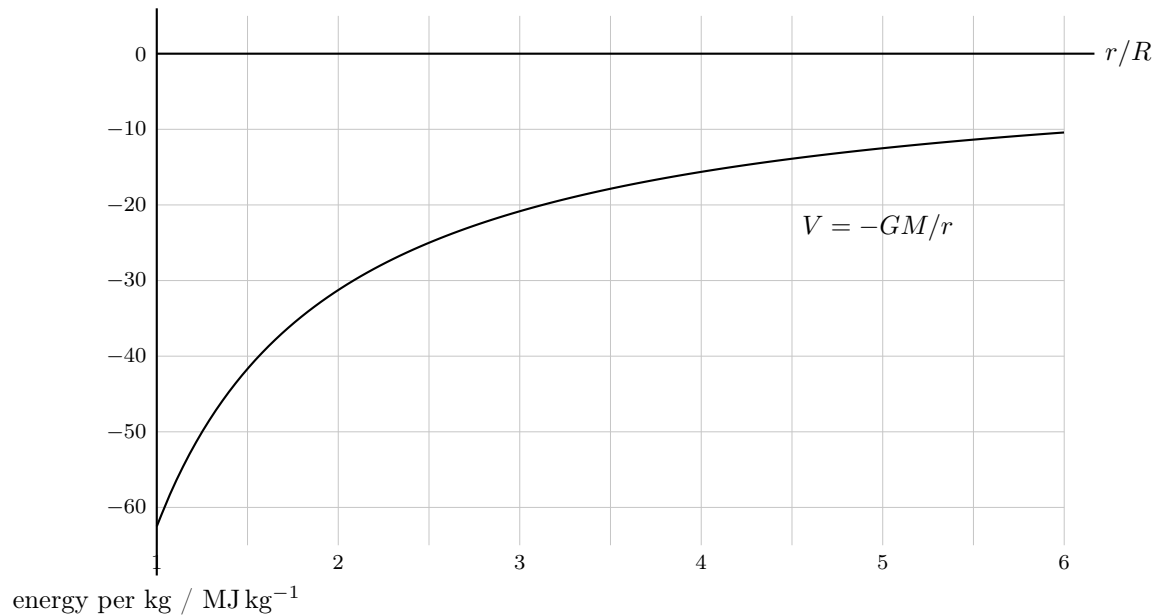
2.3. A probe of mass 1000 kg is raised from the surface to a height $h = R$. Calculate the work done (a) using mgh with $g = 9.81\text{ N kg}^{-1}$, and (b) using $m\Delta V$. [4]

2.4. Use your graph to explain which of the two answers in 2.3 is valid, and why the other one is too large. [2]

3. Escape and maximum height by energy

3.1. Derive the escape speed from the surface of a planet of mass M and radius R , and calculate it for the Earth. Explain why an energy argument is preferable to integrating the force. [4]

- 3.2.** A rocket is fired straight up from the surface at 8.0 km s^{-1} . Calculate its total energy per kilogram, and draw that energy as a horizontal line on the graph below. Mark the turning point, and use it to read the greatest distance from the centre of the Earth. [4]

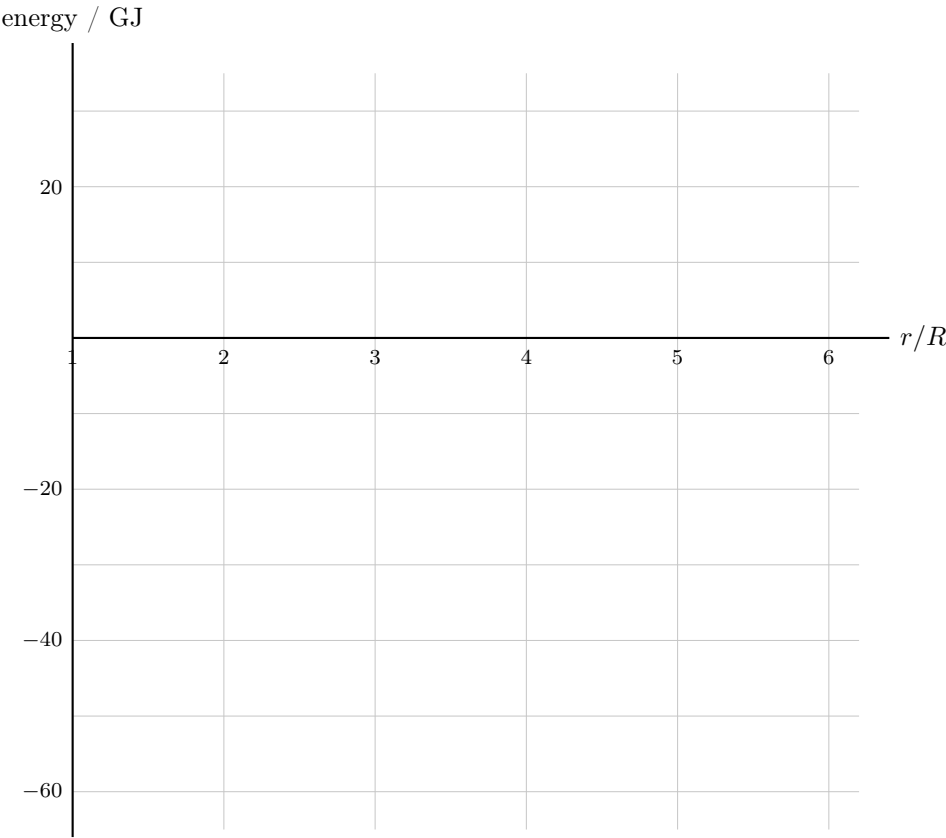


- 3.3.** The rocket is instead launched at 12.0 km s^{-1} . Calculate its speed when it is very far from the Earth. [3]

4. Circular orbits by field strength

- 4.1.** Use the force balance to show that the orbital speed at radius r is $v = \sqrt{GM/r}$, and hence that the total energy of the satellite is $-GMm/2r$. [4]

4.2. On the axes below, sketch and label the kinetic energy, the potential energy and the total energy of a 1000 kg satellite against orbital radius, and state what the zero line represents. [4]



4.3. Complete the table for two circular orbits of a 1000 kg satellite. [6]

Orbit	$r \text{ / m}$	$v \text{ / km s}^{-1}$	T	KE / GJ	PE / GJ	$E \text{ / GJ}$
Low, 400 km altitude						
Geostationary						

4.4. Air resistance causes a satellite to spiral down from an orbit of altitude 400 km to one of altitude 300 km. State and explain what happens to its speed, kinetic energy, potential energy and total energy. [5]

5. Putting into orbit and changing orbit

- 5.1.** A 1000 kg satellite is lifted from rest on the surface to a circular orbit of radius 6.77×10^6 m. Calculate the increase in potential energy, the kinetic energy it needs in orbit, and the total energy that the launch must supply. Comment on how these two contributions compare. [5]
- 5.2.** The satellite then moves from radius 6.77×10^6 m to the geostationary radius 4.22×10^7 m. Calculate the change in (a) total energy, (b) kinetic energy and (c) potential energy. Explain why the engines do positive work although the speed decreases. [6]
- 5.3.** From the low circular orbit, calculate the extra kinetic energy and the extra speed needed to escape from the Earth, and compare the energy with the 62.5 GJ needed to escape from the surface. [4]
- 5.4. Stretch.** A Hohmann transfer between the two circular orbits uses an ellipse with semi-major axis $a = (r_1 + r_2)/2$, on which $v^2 = GM(2/r - 1/a)$. Calculate the speed change at each of the two burns and the time taken to travel along the ellipse. [5]

6. Summary

Question type	Tool	Key relation
Work between two positions		
Escape		
Circular orbit speed and period		
Energy of an orbit		
Change of orbit		
Drag on a satellite		